

Skills Summary

Composite Solids, Density and Design

Use this reference while completing your worksheet or when studying.

Skill 1 — Volume of a composite solid

Method: cut the solid into standard pieces, find each volume, then *add* pieces that are stuck on and *subtract* pieces that are drilled or scooped out.

$$V_{\text{cyl}} = \pi r^2 h \quad V_{\text{cone}} = \frac{1}{3} \pi r^2 h \quad V_{\text{sphere}} = \frac{4}{3} \pi r^3 \quad (\text{hemisphere: half of that})$$

Example: A cone of radius 2 and height 6 sits on a cylinder of radius 2 and height 5. Find the volume.

Step 1. The solid is two standard shapes joined, and neither is cut away, so the volume is a sum — find each piece and add.

Step 2. $V = \pi(2)^2(5) + \frac{1}{3}\pi(2)^2(6) = 20\pi + 8\pi = 28\pi$
 $\Rightarrow V \approx 87.96$

Try it: A cone of radius 3 and height 7 sits on a cylinder of radius 3 and height 4. Find the volume, to the nearest hundredth.

check: 20.621

Skill 2 — Surface area of a composite solid

Method: count only the faces that are actually *on the outside*. A face where two pieces join is buried and is never included, and an open top means no lid.

cylinder wall = $2\pi rh$ circle = πr^2 cone curved surface = $\pi r\ell$ (slant height ℓ , not the height)

Example: An open-topped cylinder has radius 3 and height 10. Find its outside surface area.

Step 1. Only the wall and the base are real surfaces here — the top is open, so the usual second circle does not belong.

Step 2. $S = 2\pi(3)(10) + \pi(3)^2 = 60\pi + 9\pi = 69\pi$
 $\Rightarrow S \approx 216.77$

Try it: An open-topped cylinder has radius 4 and height 5. Find its outside surface area, to the nearest hundredth.

check: 86.921

Skill 3 — Density, mass and volume

One relationship, three forms: density = $\frac{\text{mass}}{\text{volume}}$, so mass = volume $\times$ density and volume = mass $\div$ density.

Check the units before you compute: g and cm^3 go together, kg and m^3 go together. A mass in kilograms against a density in g/cm^3 needs converting first.

Example: A casting of volume 350 cm^3 has density 2.7 g/cm^3 . Find its mass.

Step 1. Volume and density are known and mass is wanted, so the relationship is used in its multiplying form — density counts grams per cubic centimetre, so multiply.

Step 2. $\text{mass} = 350 \times 2.7$

$$\Rightarrow 945 \text{ g}$$

Try it: An object has mass 3200 g and density 1.6 g/cm^3 . Find its volume.

check: 2000

Skill 4 — Designing to a constraint

Method: write the formula for the quantity that is fixed, put the required value on one side, and solve for the dimension you are free to choose.

For a cylinder of fixed capacity V and fixed radius r : $h = \frac{V}{\pi r^2}$. Changing r therefore changes h too — the two are never independent once the capacity is fixed.

Example: A cylindrical tank of radius 2 must hold 40 cubic units. Find the height.

Step 1. The capacity is the fixed requirement and the radius is already chosen, so the height is the only unknown — rearrange the volume formula for it.

Step 2. $h = \frac{40}{\pi(2)^2} = \frac{40}{4\pi}$

$$\Rightarrow h \approx 3.18$$

Try it: A cylindrical tank of radius 2.5 must hold 50 cubic units. Find the height, to the nearest hundredth.

check: 2.55